

AI- and IoT-Based Early Dengue Prediction and Prevention System

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Abstract—Dengue fever remains a persistent public health challenge in densely populated urban regions due to favorable environmental conditions for mosquito breeding. This paper presents an AI- and IoT-based dengue prediction and prevention system that integrates real-time environmental sensing with machine-learning-driven risk assessment. A low-cost embedded prototype is implemented using an ESP32 microcontroller with temperature, humidity, water-level, and infrared sensing modules to monitor mosquito breeding indicators and activity. Sensor data are transmitted to a cloud platform and analyzed using a hybrid XGBoost–LSTM model to estimate dengue risk probabilities. Model evaluation was conducted using supervised learning with an 80:20 training–testing split, and performance stability was assessed using MAE and RMSE metrics. Scenario-based impact analysis demonstrates that timely detection and preventive intervention can progressively reduce model-estimated dengue risk under defined assumptions. Experimental prototype validation using the Blynk IoT platform confirms stable data acquisition, real-time visualization, and alert generation. The proposed system demonstrates practical feasibility and scalability for urban dengue risk monitoring and future smart city deployments. The primary contribution of this work lies in demonstrating a deployable AI–IoT framework for proactive dengue risk monitoring in resource-constrained urban environments.

Index Terms—Dengue prediction, Internet of Things (IoT), ESP32, environmental sensing, mosquito detection, machine learning, XGBoost, LSTM, smart healthcare, urban disease monitoring

I. INTRODUCTION

Dengue is one of the most significant mosquito-borne diseases affecting tropical and subtropical regions [1]. In Bangladesh, increasing temperature, humidity, and rainfall patterns, together with dense urbanization, contribute to recurrent dengue outbreaks [11], [12]. Existing control practices are often reactive and depend on delayed reporting, which reduces intervention effectiveness [11]. This work addresses the need for proactive, low-cost monitoring by combining IoT sensing with AI-based risk estimation.

A. Dataset Description

The dataset used in this study was obtained from a publicly available Kaggle repository titled “Dengue Dataset — Bangladesh”, curated by Kawsar Ahmad and Farzana Eva. Historical dengue and survey-based diagnostic data were obtained from a publicly available Kaggle dataset titled “Dengue Dataset — Bangladesh”, curated by Kawsar Ahmad and

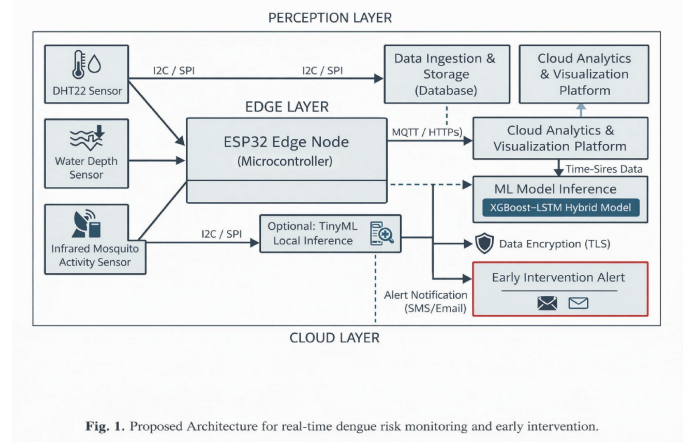


Fig. 1. Proposed Architecture for real-time dengue risk monitoring and early intervention.

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Farzana Eva [16]. The dataset reports temporal records starting from September 1, 2023, and includes respondent-level demographic attributes, diagnostic test indicators, and location-related fields within the Dhaka region.

Key features used in this study include:

- **Gender:** Gender of the survey respondent.
- **Age:** Age of the survey respondent.
- **NS1 (Non-Structural Protein 1):** Binary diagnostic test result, where 1 indicates positive and 0 indicates negative.
- **IgG (Immunoglobulin G):** Binary antibody test result, where 1 indicates positive and 0 indicates negative.
- **IgM (Immunoglobulin M):** Binary antibody test result, where 1 indicates positive and 0 indicates negative.
- **Area:** Geographic area within the Dhaka region where the respondent resides.
- **Area Type:** Classification of the area as *Developed* or *Undeveloped*.
- **House Type:** Type of housing, such as *Building*, *Tin-Shed*, or *Others*.
- **District:** Administrative district within the Dhaka region.
- **Outcome:** Target variable indicating the presence (1) or absence (0) of reported dengue cases.

For model development, the dataset was cleaned and encoded where necessary, then divided into training and testing

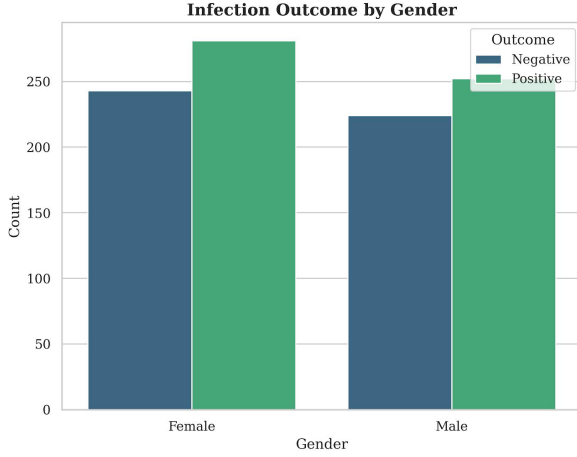


Fig. 2. Distribution of dengue infection outcomes by gender.

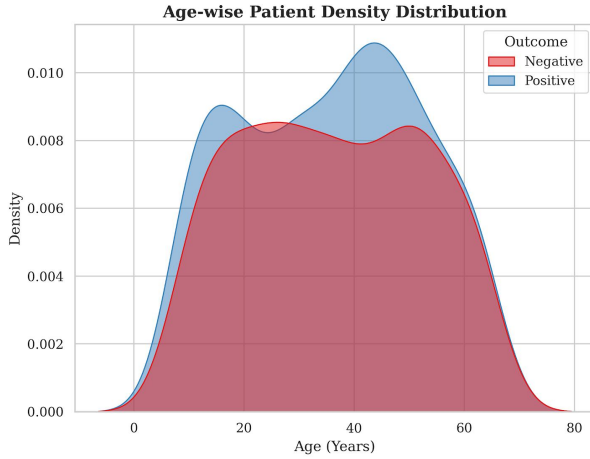


Fig. 3. Age-wise distribution of patients in the dataset.

subsets using an 80:20 ratio to enable supervised learning and evaluation of the proposed hybrid model.

The exploratory analysis results shown in Figs. 2–9 reveal demographic, environmental, and clinical patterns that guide feature engineering for the proposed dengue risk prediction model. The dataset distributions are summarized using exploratory visualizations (Figs. 2–9), highlighting demographic, clinical, and location-based patterns that motivate feature selection for the proposed risk model.

B. Risk Modeling

The dengue risk estimation engine is implemented using a hybrid XGBoost–LSTM architecture. Temporal sequences of environmental sensor data are processed by the LSTM component to capture time-dependent patterns, while aggregated feature representations are evaluated using XGBoost to model non-linear relationships. Due to the ensemble and recurrent

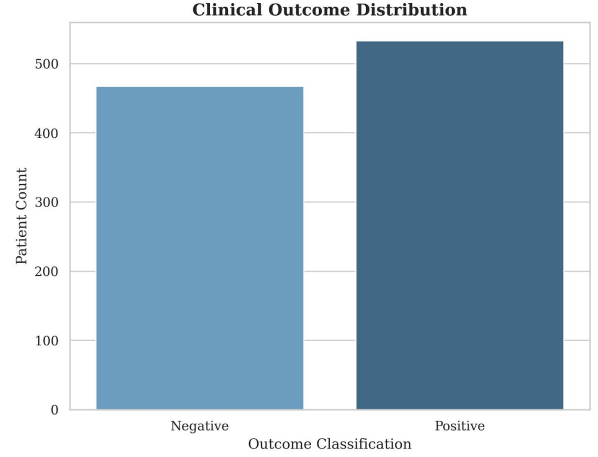


Fig. 4. Overall distribution of dengue outcomes in the dataset.

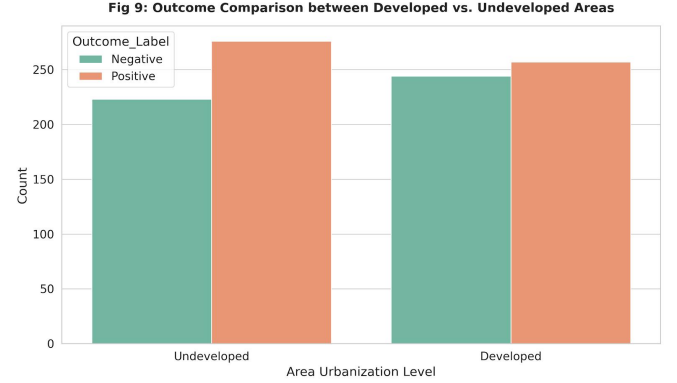


Fig. 5. Comparison of dengue outcomes between developed and undeveloped areas.

nature of the model, a closed-form analytical expression of the inference process is not available.

For interpretability and reporting purposes, the final dengue risk score is expressed as a normalized probability:

$$R = \sigma(\hat{y}) \quad (1)$$

where $\hat{y}$ denotes the fused output of the hybrid XGBoost–LSTM model and $\sigma(\cdot)$ represents a sigmoid-based normalization function.

C. Model Training and Evaluation

The hybrid XGBoost–LSTM model was trained using time-series environmental features. The LSTM component captures temporal dependencies in sequential sensor data, while XGBoost performs gradient-boosted learning on aggregated feature representations.

The dataset was divided into 80% training and 20% testing data. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and classification accuracy of predicted risk levels. Hyperparameters were

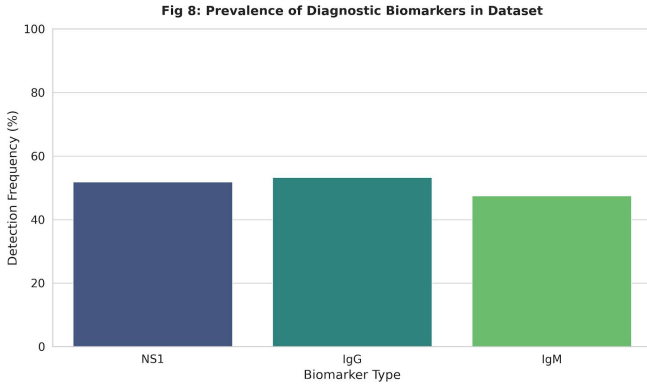


Fig. 6. Prevalence of NS1, IgG, and IgM biomarkers in the dataset.

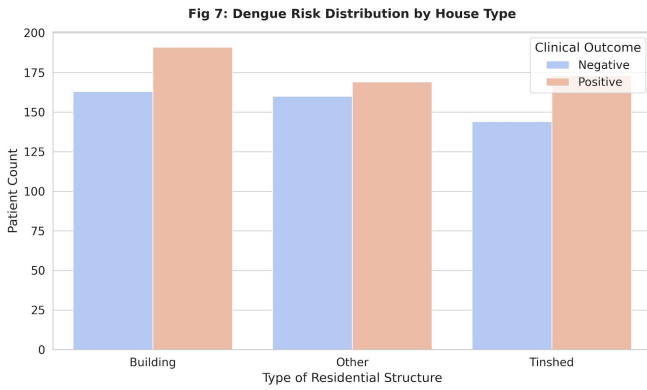


Fig. 7. Dengue risk distribution across different housing types.

tuned empirically to balance model complexity and generalization performance. The hybrid architecture improves predictive stability by combining temporal pattern learning (LSTM) with nonlinear feature optimization (XGBoost), allowing robust modeling of complex environmental–disease relationships.

D. Mathematical Formulation of Risk Estimation

Let the environmental input sequence be:

$$X_t = [T_t, H_t, W_t, M_t]$$

where T_t is temperature, H_t is humidity, W_t denotes stagnant water indicators (e.g., water depth/standing water), and M_t represents mosquito activity counts at time t .

The LSTM processes temporal patterns:

$$h_t = \text{LSTM}(X_t, h_{t-1})$$

The extracted representation is then evaluated using XGBoost:

$$\hat{y} = f_{\text{XGBoost}}(h_t)$$

Finally, the dengue risk probability is obtained via sigmoid normalization:

$$R = \sigma(\hat{y}) = \frac{1}{1 + e^{-\hat{y}}}$$

This formulation enables dynamic mapping of environmental patterns to a normalized dengue outbreak likelihood score.

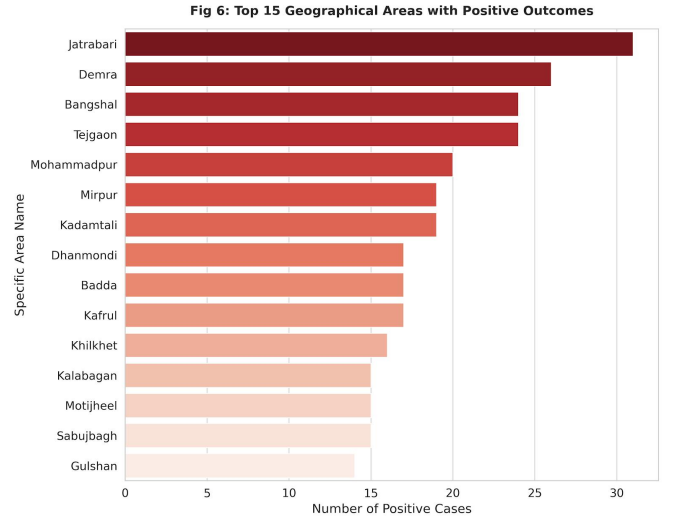


Fig. 8. Top geographical areas with the highest dengue-positive cases.

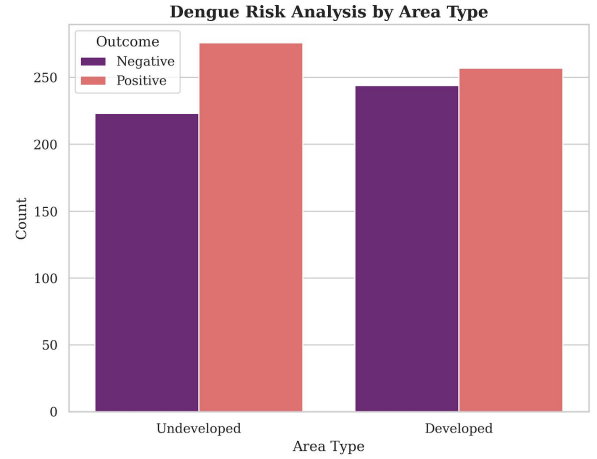


Fig. 9. Dengue risk analysis by area type.

E. Prototype Implementation and Validation

A functional prototype was implemented to validate feasibility. The prototype integrates an ESP32 microcontroller with temperature and humidity sensors, a water depth level sensor for stagnant water detection, and an infrared sensing module for mosquito presence and activity counting. Data were streamed to a cloud dashboard using the Blynk IoT platform and validated under controlled testing, confirming stable acquisition, connectivity, visualization, and alert triggering.

II. RESULTS AND DISCUSSION

A. Historical Infection Rates

Historical infection rates were summarized by zone to contextualize risk levels.

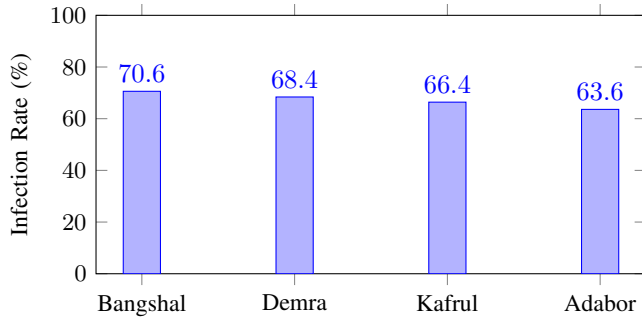


Fig. 10. Historical dengue infection rates by zone derived from the dataset used in this study [16].

TABLE I
REAL-TIME PROTOTYPE SENSOR-BASED DENGUE RISK ASSESSMENT
(MIRPUR TEST SITES)

Test Site (Mirpur)	Risk (%)	Observed Condition
Rupnagar (Near Home)	81.6	Stagnant water indicator high
Arambagh (Mirpur Area)	74.0	Elevated heat and humidity
Mirpur (Controlled Indoor)	12.0	Controlled environment

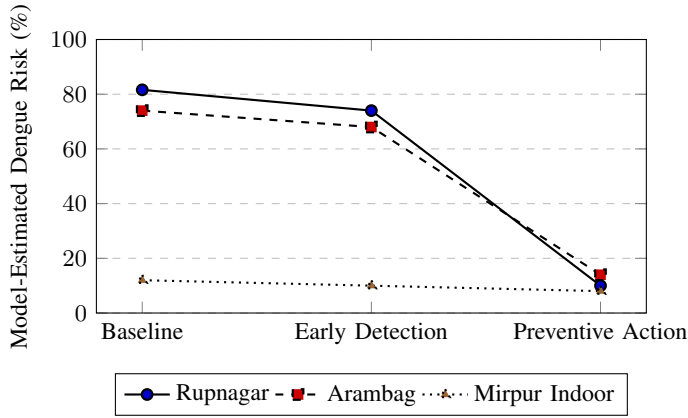


Fig. 11. Multi-zone scenario-based impact analysis of model-estimated dengue risk reduction across intervention stages.

B. Real-Time Sensor-Based Risk Assessment

The trained model was applied to real-time sensor inputs collected through the prototype. The reported values represent model-estimated probabilities rather than confirmed clinical infection rates.

The values in Table I were obtained from prototype testing at the listed Mirpur locations and represent model-estimated risk levels rather than confirmed clinical infection rates.

C. Impact Analysis

Impact analysis evaluates progressive reduction of model-estimated risk under staged intervention assumptions (early detection and preventive action). These results represent modeled intervention response rather than confirmed epidemiological reduction and are intended to illustrate potential preventive effectiveness under controlled assumptions. The multi-zone presentation improves generality across urban contexts.

III. FUTURE WORK AND SYSTEM SCALABILITY

Future work will focus on city-scale deployment, edge-level inference to reduce latency, continuous model retraining using field data, and strengthened security through device authentication and encrypted communication [13]. Long-term deployment will enable longitudinal evaluation aligned with public health programs in Bangladesh [11]. Future enhancements will include secure device authentication, encrypted data transmission (e.g., TLS), and cloud-side access control to protect sensitive health-related data and improve resilience against cyber threats in large-scale smart city deployments.

IV. CONCLUSION

This paper presented an AI- and IoT-based dengue prediction and prevention system integrating real-time sensing, prototype validation, and machine-learning-driven risk estimation. The ESP32-based prototype demonstrated stable sensing and cloud integration using the Blynk platform. Scenario-based multi-zone impact analysis shows that timely detection and preventive intervention can reduce model-estimated risk under defined assumptions. This study contributes to the field by bridging environmental sensing with predictive public health analytics through a practical embedded AI-IoT framework. The framework provides a practical foundation for scalable dengue surveillance and smart healthcare deployments.

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